

Back

confidential do not share

or else

u gay

Answer

Reset

Hide

1. What is not true about concurrency?

Operating system can run many processes at the same time on multi-core processor

Operating system can run many threads at the same time on multi-core processor

Memory of a process cannot be shared with other processes

When a process has several threads, race condition might occur

Only one process can be executed on a core of a processor at one time

2. What is not true about thread?

Each thread has separate address spaces

Each thread has its own PC

All threads share resource within a process

OS also needs to context-switch threads

Thread shares data but has separate stack

3. What is not true about race condition?

To prevent race condition, mutex must be guaranteed when critical region is entered

Mutex allows one of several threads to enter a critical region

Race condition makes process runs faster

Race condition might occur when threads share resources

Race condition will not occur if one thread is running at a time

4. Which statement cannot be a critical region?

```
for(int i=10;i>0;i--);
```

```
count = count + 1
```

```
sum += i
```

```
i++
```

all of them can be a critical region

How the Race Condition Occurs

The high-level programming instruction `sum += 1` is typically broken down into multiple low-level assembly instructions by the compiler.

Read: The current value of `sum` is read from memory into a CPU register.

Modify: The value in the register is incremented by one.

Write: The new value from the register is written back to the memory location of `sum`.

```
mov 0x8049a1c, %eax
add $0x1, %eax
mov %eax, 0x8049a1c
```

5. Which system call cause the main thread to wait for child thread to terminate?

`pthread_cond_wait`

`pthread_join`

`pthread_create`

`pthread_cond_init`

`pthread_mutex_init`

6. What is not true about lock?

compare_and_swap is a hardware lock

Spin lock wastes CPU cycles waiting

Software test and set is not correct

We can create pure software lock

We can you interrupt to create lock

7. What is true about lock?

Fetch and add does not know the order of lock request

Compare-and-swap is fairer than test-and-set

Process can only wait for lock by spinning

Load-linked allows caller to load a ticket

Starvation might happen when uses only test-and-set

8. What is the software not a support to solve wasting cycles with lock?

Sleep instead of spin

Yield system call

Park when lock

Using waiting queue for lock request

All are software supports to solve wasting cycles

Producers/Consumers

9. What is not true about bounded buffer problem?

The program does not work with one conditional variable

Producer is waiting on empty condition

Consumer is waiting on full condition

Mutex is used to count available items

You need all mutex, full, and empty

10. . What does a producer call when it produces something?

mutex_lock(&mutex)

mutex_unlock(&mutex)

cond_signal(&fill)

cond_signal(&empty)

`cond_wait(&fill,&mutex)`

11. What does a consumer call when it consume something?

`mutex_lock(&mutex)`

`mutex_unlock(&mutex)`

`cond_signal(&fill)`

`cond_signal(&empty)`

`cond_wait(&fill,&mutex)`

12. What is true about semaphore?

There are binary and continuous semaphore

When `sem_post` is called, value is decreased by 1

When `sem_wait` is call, value is increased by 1

Semaphore value can only be 0 and 1

Semaphore can be used to create mutex *by initializing it to 1*

13. Which statement is not true?

Each resource types only have one instance *ଏକক*

A process has to request for a resource before use

A process needs to hold a resource while using it

When a process finish the task, resources can be released

A process cannot hold processor

14. Which problem does cause by resource allocation?

Race condition

Deadlock

Trashing

Internal fragmentation

Zombie

15. What is true about deadlock?

A deadlock needs more than one process to occur

Deadlock will occur when many processes are waiting for CPU

Deadlock will occur when virtual memory is running low

Deadlock will occur if process need to hold resources and wait for other resources

Deadlock will occur even there is only one resource type

16. What is not the four conditions of deadlock?

Mutual exclusion

Non-preemption

Virtualization

Hold and wait

Circular wait

17. When does a deadlock might occur?

Resource allocation graph has circle

Banker algorithm is in a safe state

There is no resource in holding

Resource can be preempted

Resource allocation graph has no circle

18. A deadlock will never occur with which deadlock management

Prevention

Avoidance

Detection

Both prevention and avoidance

Deadlock will occur in all
management

19. Which deadlock management needs recovery?

Prevention

Avoidance

Detection

Both prevention and avoidance

All managements need recovery
scheme

20. Which problem does requesting resource in increasing order would solve?

No preemption

Hold and wait

Mutual exclusion

Circular wait

Starvation

21. Which method can solve no preemption?

test_and_set

yield

try_lock and try_wait

fetch_and_add

request resource in increasing order

22. What is not true about Banker's algorithm?

Process must declare maximum number of resources

Deadlock might occur in safe state

Need is Max minus allocation

Deadlock will occur in unsafe state

Request can be granted if resource left is more than needed resource

23. What could deadlock recovery do?

Deny request

Terminate processes

Preempt resource from process

Put process on wait queue

Put process on ready queue

24. Which statement is not true?

Wait-for graph is equivalent to resource allocation graph

Resource allocation graph is used in deadlock avoidance

Resource allocation graph is used in deadlock detection

There is no resource in wait-for graph

Deadlock occurs when there is a loop in wait-for graph

25. Which method is when the device-registers are mapped into the memory space of the

System call

Interrupt

Direct memory access

Memory-Mapped I/O

Pooling

26. Which statement is not true?

Pooling is the most efficient method to access IO

Interrupt can be hardware and software

Interrupt frees up processor to do something else

DMA moves data from device directly into memory

Interrupt vector table stores locations of service routines

27. Which layer knows how to read and write specific block of device?

Application layer

File system

Generic block layer

Device driver

All layers can directly read and write blocks on device

28. Which Disk Scheduling Algorithms might cause starvation?

FCFS

SSTF *သိသည့်နေရာမှ current location မှာပါရှိသော*

SCAN

LOOK

No starvation in disk scheduling

29. Which Disk Scheduling Algorithms does sweeps from the outer to inner track and then back to the outer, again?

FCFS

SSTF

SPTF

SCAN

LOOK

30. What is not true about RAID

RAID 0 is stripping with no mirror

RAID 1 is mirror with no stripping

RAID 2 has error correcting code

RAID 3 is Bit-striping and dedicated parity disk

RAID 4 is Block-striping and block-mirroring

31. Which file allocation method is FAT?

Linked allocation

Contiguous allocation

Indexed allocation

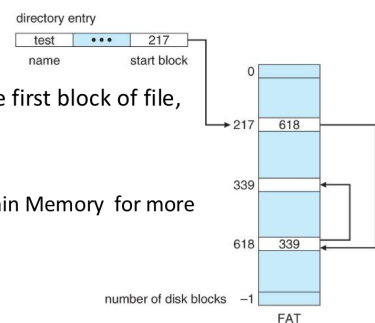
Multi-level indexed allocation

Combined scheme

Linked Allocation (2)

- eg. flash 32 18min Link
• The File Allocation Table (FAT) is a variation of linked allocation, where all the links are stored in a separate table at the beginning of the partition

- The directory contains the first block of file, used to index the FAT
- Advantage
 - FAT can be cached in Main Memory for more efficiency



CPE 326 Operating System Final Exam

Part 2-2

How can we prevent deadlock in dining philosopher problem? Explain in detail.

เราจะแก้ไขปัญหของ dining philosopher ด้วยการกำหนดให้ philosopher ที่เป็นเลขคู่ถือ fork ด้านขวาและรอ fork ทางด้านซ้าย ส่วน philosopher ที่เป็นเลขคี่ถือ fork ด้านซ้ายแล้วรอ fork ทางด้านขวา

```
void get fork(p) {
    If (p % 2 == 0){ // philosopher เลขคู่
        Fork(right(p))
        Fork(left(p))
    }
    Else {           // philosopher เลขคี่
        Fork(left(p))
        Fork(right(p))
    }
}
```

ทำให้ philosopher จะไม่ต้องการอันที่ตรงกันในจังหวะเดียวกัน

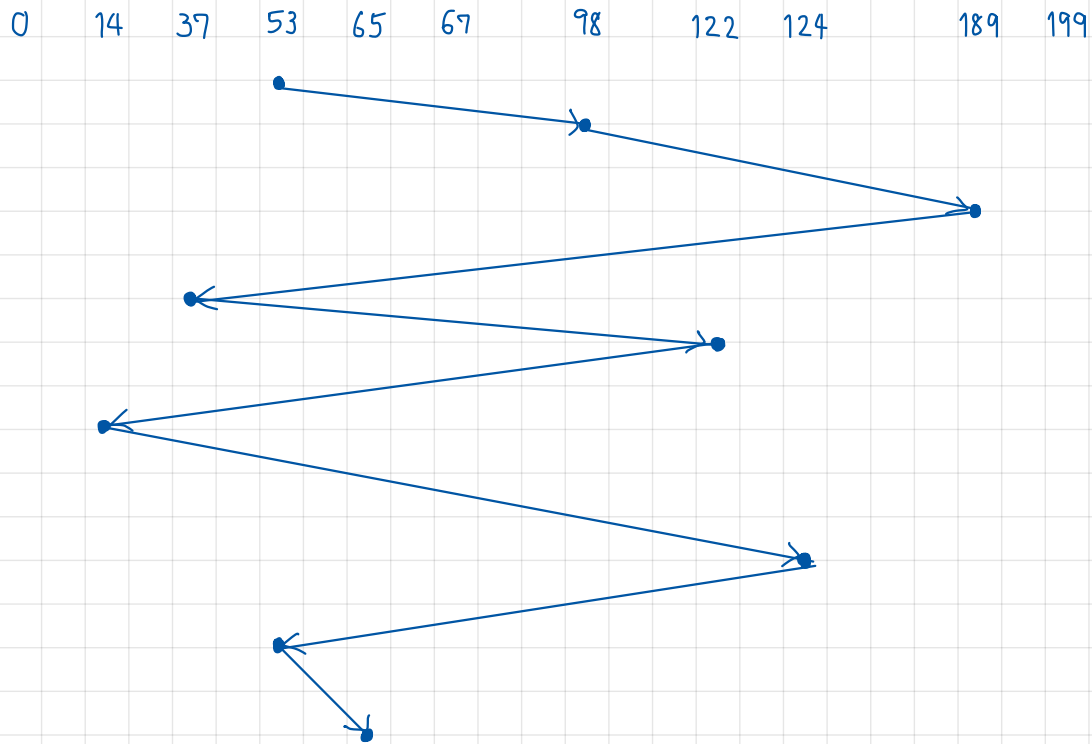
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Part2-3

If a disk has cylinder 0-199, and a head starts at position 53. The following positions are in reading queue {98,189,37,122,14,124,65,67}. Illustrate the seek path and calculate the total head movement for the following algorithms:

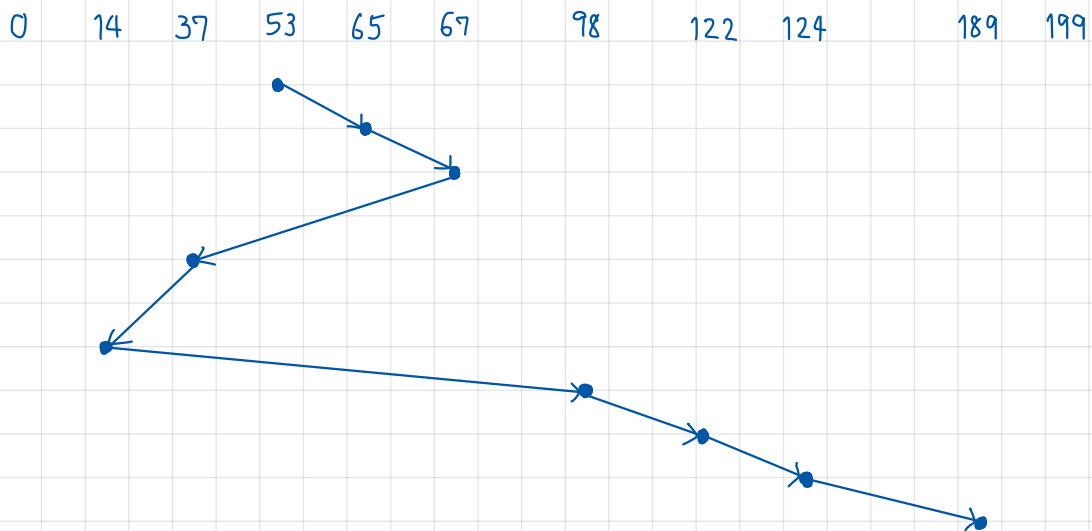
1. FCFS
2. SSTF
3. C-SCAN (left to right)
4. C-LOOK (left to right)

1. FCFS



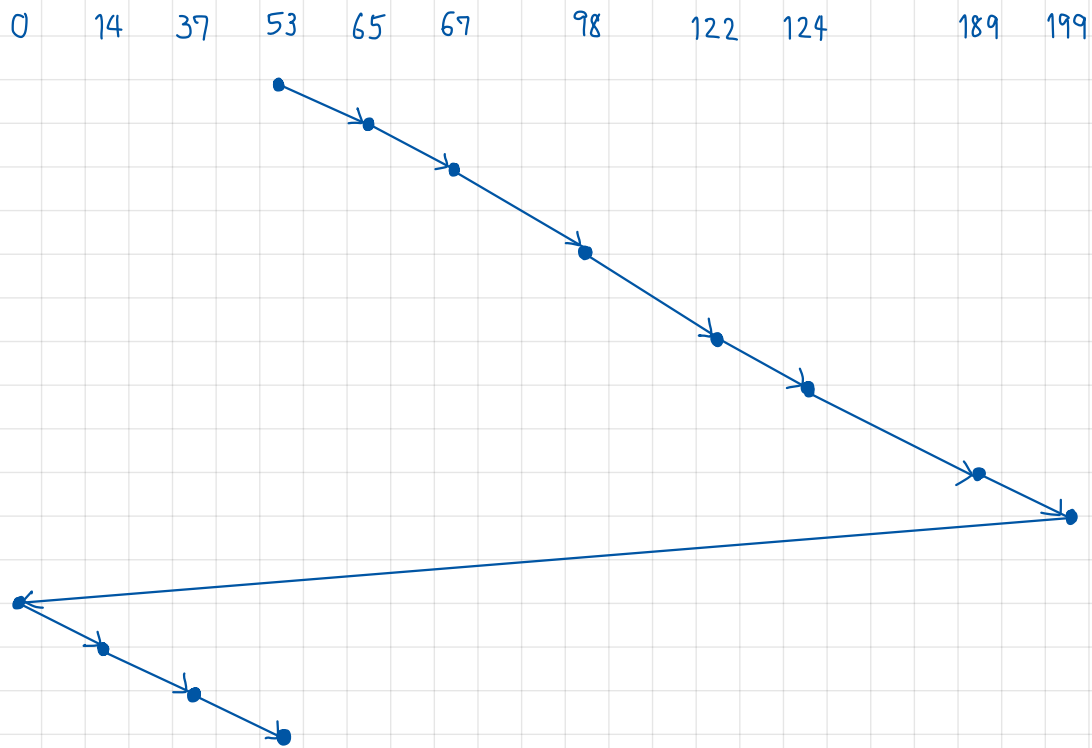
total head movement = 652

2. SSTF



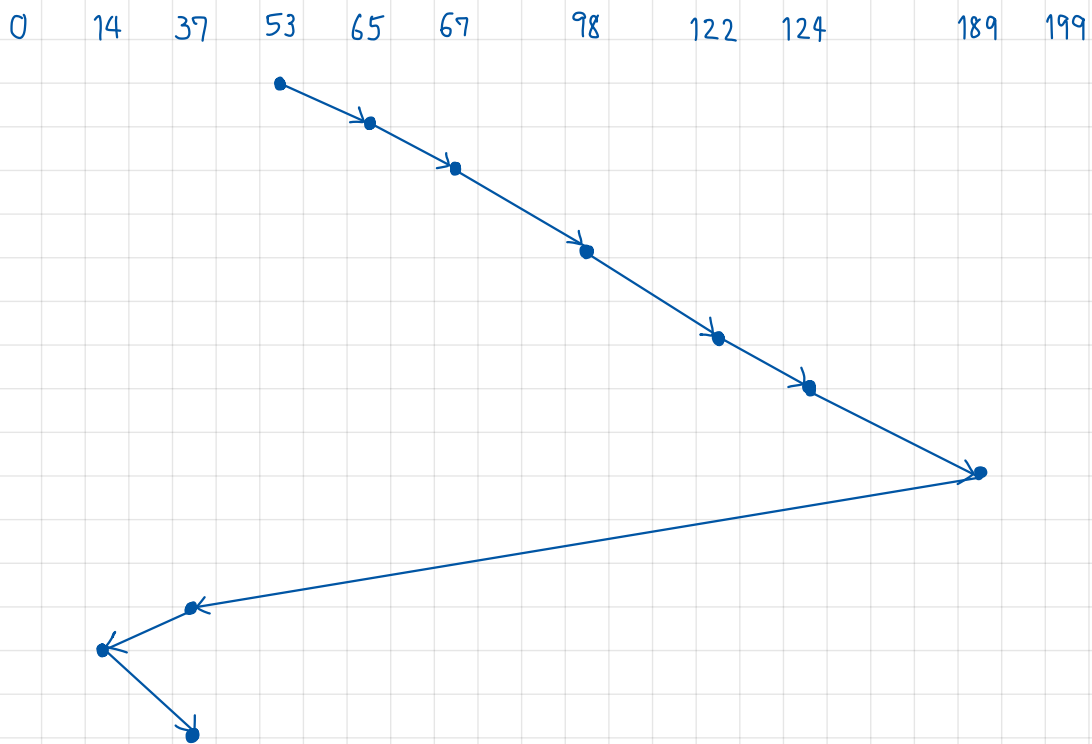
total head movement = 242

3. C-SCAN



total head movement = 382

4. C-LOOK



total head movement = 334

Part2-1

1. What is the benefit of thread over process when context switching occurs?
- Thread process switching will keep virtual memory space the same while switching
2. Explain how can a race condition effect the result of concurrency execution?
- Race condition จะเกิดขึ้นเมื่อหลาย process เข้าถึง resource ในเวลาเดียวกัน

P1	P2
Read $i + 1$	prepare read i
Print i	$i + 2$ Print i

3. Give an example of code to create a lock from test_and_set()

```
int TestAndSet(int *ptr, int new) {
    int old = *ptr; // fetch old value at ptr
    *ptr = new;     // store 'new' into ptr
    return old;     // return the old value
}
```

```
typedef struct __lock_t {
    int flag;
} lock_t;
```

```
void init(lock_t *lock) {
    // 0 indicates lock is available, 1 that it is held
    lock->flag = 0;
}
```

```
void lock(lock_t *lock) {
    while (TestAndSet(&lock->flag, 1) == 1); // spin-lock
}
```

Test-and-set (atomic instruction)

```
void unlock(lock_t *lock) {
    lock->flag = 0;
}
```

This is called "spin lock"
spins using CPU cycles,
until the lock becomes
available.

4. How can you eliminate the time wasted in spinning loop in lock?
- Yield แล้วไปทำอย่างอื่นแทน